8p Inverted Duplication/Deletion Syndrome (inv dup del(8p)) — Comprehensive Disease Characterization Report

Disease: 8p inverted duplication/deletion syndrome **MONDO ID:** MONDO:0019876 **Orphanet:** ORPHA:96092 ("Chromosome 8p inverted duplication/deletion") **Category:** Chromosomal (contiguous-gene) disorder **Report basis:** 5 autonomous investigation iterations, 11 confirmed findings, 53 primary papers reviewed

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Summary

8p inverted duplication/deletion syndrome (**inv dup del(8p)**) is a rare, almost invariably sporadic (de novo) contiguous-gene chromosomal disorder. It is defined by a single rearranged chromosome 8 that simultaneously carries a **terminal 8p deletion** (~6.7–11.4 Mb, 8p23.3→8p23.1) and an **inverted interstitial duplication** (~30–32 Mb, 8p23.1→8p11.1), the two imbalances being separated by a preserved single-copy (disomic) region delimited by the olfactory-receptor (OR) low-copy-repeat clusters (REPD/REPP). The rearrangement arises predominantly by **non-allelic homologous recombination (NAHR)** during **maternal meiosis** between these OR repeats, an event strongly predisposed by a common **paracentric 8p23.1 inversion polymorphism** carried in the heterozygous state by ~26% of individuals of European descent. Because the operative risk factor is the mother's benign inversion-carrier status rather than maternal age, and because the event is de novo, recurrence risk in a family is low.

The dual dosage imbalance drives a highly penetrant, multisystem phenotype. Terminal haploinsufficiency of dosage-sensitive transcription factors — most importantly **GATA4** and **SOX7** at 8p23.1 — together with triplosensitivity of the large interstitial duplication produces a recognizable clinical picture: developmental delay / intellectual disability in ~97% of patients, anomalies of the corpus callosum (agenesis/hypoplasia) in ~60–65%, infantile hypotonia,

characteristic facial dysmorphism, congenital heart defects (notably atrioventricular/atrial septal defects), and seizures in ~30–35% (mean onset ~3.9 years). Severity is graded by the size and breakpoints of the imbalance and is attenuated by somatic mosaicism. Model organisms (mouse *Gata4*, zebrafish *sox7*) recapitulate the cardiac and vascular components, validating the dosage mechanism for the heart phenotype, though no model reproduces the whole contiguous imbalance.

Diagnosis is molecular-cytogenetic: **chromosomal microarray (CMA/aCGH/SNP-array)** defines the deletion+duplication pattern with precise breakpoints, with **karyotype** (typically add(8)(p23)/der(8)) and **FISH** confirmation and parental karyotyping to document de novo origin. Prenatally, diagnosis is triggered by ultrasound anomalies (increased nuchal translucency, ventriculomegaly, cardiac and renal defects) that can mimic trisomy 18. There is **no curative or disease-specific therapy**; management is symptomatic and multidisciplinary, and prevention is limited to genetic counseling and prenatal/preimplantation diagnosis.

Section 1 — Disease Information

Overview. inv dup del(8p) is a recurrent structural chromosomal rearrangement of the short arm of chromosome 8, producing a contiguous-gene syndrome from combined deletion (dosage loss) and inverted duplication (dosage gain). The clinical entity is dominated by neurodevelopmental impairment, brain (corpus callosum) malformation, hypotonia, dysmorphism, and congenital heart disease.

Key identifiers. - **MONDO:** MONDO:0019876 - **Orphanet:** ORPHA:96092 ("Chromosome 8p inverted duplication/deletion") - **MeSH:** concept mapped to *Chromosomes, Human, Pair 8* - **OMIM:** No single dedicated phenotype MIM number; covered under the contiguous-gene 8p23.1 deletion/duplication entries and **GATA4** (%607941) - **ICD-10:** Q99.9 / Q93–Q95 range (chromosomal abnormalities); ICD-11: LD44 (chromosomal anomalies)

Synonyms / alternative names: inv dup del(8p); invdupdel(8p); inverted duplication deletion 8p syndrome; inverted duplication of 8p with terminal deletion; der(8) inverted duplication deletion syndrome; recombinant chromosome 8 (partial).

Information source. The knowledge base for this disorder is derived from **aggregated individual case reports and small cohort/case series** (the largest being 36 new patients plus a literature review), supplemented by prenatal diagnostic series and disease-level cytogenetic resources (Orphanet, DECIPHER, ClinVar). There is no large population-level EHR dataset.

Section 2 — Etiology

Primary cause (genetic, structural). The disorder is caused by a de novo structural chromosomal rearrangement, not by environmental or infectious factors. The recurrent rearrangement combines a distal terminal deletion (8p23.3→8p23.1) with an inverted interstitial duplication (8p23.1→8p11.1) separated by a disomic region delimited by the OR gene clusters.

Mechanism. The rearrangement arises by **NAHR during maternal meiosis** between segmental duplications made up of OR gene clusters (REPD/REPP), facilitated by a paracentric **8p23.1 inversion polymorphism** present in ~26% of Europeans. A dicentric chromosome intermediate forms, breaks, and the resulting terminal deletion is stabilized by **telomere healing** (direct addition of telomeric repeats), or rarely by **telomere capture from 8q**.

> "Inverted 8p duplication deletions are recurrent chromosomal rearrangements that most often arise through non-allelic homologous recombination (NAHR) during maternal meiosis between segmental duplications made up of the olfactory receptor (OR) gene clusters. The presence of a paracentric inversion polymorphism in 8p23.1, found in approximately 26% of European population, may trigger meiotic misalignment and NAHR between the OR gene repeats." — PMID: 24502041
>
> "The terminal deletions are stabilized by direct addition of telomeric repeats, so called telomere healing." — PMID: 19041960

Genetic risk factor. The single established predisposing factor is **maternal heterozygosity for the 8p23.1 inversion polymorphism**, a benign structural variant that itself causes no phenotype but confers susceptibility to meiotic malsegregation. In the landmark study all 8 mothers of inv dup(8p) probands were inversion carriers (PMID: 11231899).

Environmental risk factors / protective factors / gene-environment interactions. None are established. Because the event is a meiotic recombination error, there are no known environmental risk factors, protective exposures, or gene–environment interactions. Unlike aneuploidies, **maternal age is not the operative factor**; maternal inversion-carrier status is.

Section 3 — Phenotypes

The phenotype is multisystem and highly penetrant. Frequencies below are drawn primarily from the largest cohort (36 new patients + literature review; PMID: 34866188) and independent case series (PMID: 35327368).

Phenotype	Type	Frequency	Onset	HPO suggestion
Developmental delay / intellectual disability	Neurodevelopmental	~97% (32/33)	Neonatal–infancy	HP:0001263 / HP:0001249
Anomalies of the corpus callosum (agenesis/ hypoplasia)	Structural CNS	~63% (17/27 imaged)	Congenital	HP:0001274 / HP:0001338
Muscular hypotonia	Neuromuscular	Very frequent (near-universal in series)	Neonatal	HP:0001252
Dysmorphic facial features	Physical	Present in all cases of one series	Congenital	HP:0001999
Seizures / epilepsy	Neurological	~34%	Mean 3.9 y (2 mo–9 y)	HP:0001250
Congenital heart defects (ASD/AVSD, others)	Cardiovascular	Frequent	Congenital	HP:0001627 / HP:0001671
Psychomotor and language delay	Neurodevelopmental	Very frequent	Infancy	HP:0011342 / HP:0000750
Orthopedic/skeletal anomalies (scoliosis, limb/joint)	Musculoskeletal	Frequent	Childhood	HP:0002751 / HP:0002650
Microcephaly	Physical/CNS	Variable (deletion component)	Congenital	HP:0000252
Behavioral problems	Behavioral	Variable	Childhood	HP:0000708

Prenatal presentation. Can mimic trisomy 18: increased nuchal translucency, ventriculomegaly, cardiac defects (including hypoplastic left heart), renal anomalies, and craniofacial dysmorphism.

Characteristics. Onset is congenital/neonatal; severity is variable (mild to severe); course is generally **stable/non-progressive** structurally but with lifelong disability; seizures may be episodic. Quality-of-life impact is substantial and lifelong: most survivors have persistent intellectual disability, hypotonia, and require ongoing developmental support, affecting mobility, communication, self-care, and independent living.

> "97% (n = 32/33) of patients presented with mild to severe developmental delay/ID and 34% had seizures with mean age of onset of 3.9 years (2 months-9 years). Moreover, out of the 24 patients with brain MRI and 3 fetuses with neuropathology analysis, 63% (n = 17/27) had AnCC." — PMID: 34866188
>
> "The main clinical manifestations in all cases are psychomotor and language delay, muscle hypotonia, and dysmorphic facial features. Malformations of the central nervous system, such as corpus callosum agenesis, were found in five cases." — PMID: 35327368

Section 4 — Genetic / Molecular Information

Causal lesion. A single derivative chromosome 8 carrying a terminal deletion (8p23.3→8p23.1) + an inverted interstitial duplication (8p23.1→8p11.1), separated by a disomic segment bounded by OR clusters. This is a **structural (copy-number) variant**, classified **pathogenic** per ACMG/AMP CNV guidelines given established dosage-sensitive gene content.

Key dosage-sensitive genes. - **GATA4** (8p23.1; HGNC:4173; OMIM %607941) — cardiac transcription factor; haploinsufficiency in the deletion drives congenital heart defects and contributes to diaphragmatic hernia. - **SOX7** (8p23.1; HGNC:18196) — SoxF transcription factor; deletion contributes to cardiac/vascular defects. - **NEIL2** (8p23.1) — implicated with GATA4/SOX7 in diaphragmatic defects by protein-interaction network analysis. - **Defensin (DEFA/DEFB) cluster** (8p23.1) — deletion associated with reduced NK-cell activity and low α-defensin, contributing to infectious vulnerability. - Candidate genes **XKR6** and **MIR597** proposed for absence seizures within the 8p23.1 interval.

> "implicated GATA4, NEIL2, and SOX7 in diaphragmatic defects. Sequence analysis of these genes in 226 chromosomally normal CDH patients, as well as in a small number of deletion 8p23.1 patients, showed rare unreported variants in the coding region" — PMID: 23165946
>
> "This patient showed lower NK cell activity and α-defensin level compared with healthy controls. These results suggest that decreased NK cell activity can result from DEF haploinsufficiency." — PMID: 35768224

Reciprocal / related conditions. The same NAHR mechanism generates reciprocal products: the supernumerary +der(8)(8p23.1pter) marker, the recurrent 8p23.1 interstitial deletion syndrome, and isolated 8p23.1 duplication syndrome (prevalence ~1/58,000), which produces a milder overlapping phenotype with GATA4 gain also linked to CHD.

> "The 8p23.1 duplication syndrome (8p23.1 DS) is a recurrent genomic condition with an estimated prevalence of 1 in 58,000." — PMID: 26097203

Modifier genes / epigenetics. No specific trans-acting modifier genes are established for inv dup del(8p); phenotype modulation is driven mainly by rearrangement size/breakpoints and mosaicism (Section 9). No disease-specific epigenetic signature is documented.

Chromosomal abnormality classification. Complex intrachromosomal rearrangement (terminal deletion + inverted interstitial duplication) — cytogenetically add(8)(p23)/der(8). Allele frequencies are not applicable (recurrent de novo structural event, not an SNV). Origin is **germline** (maternal meiosis); somatic (postzygotic) mosaic forms occur.

Section 5 — Environmental Information

Not applicable. No environmental factors, lifestyle factors, or infectious agents are implicated in causing inv dup del(8p). The disorder is a de novo meiotic recombination error. (Affected individuals have increased vulnerability to infections as a downstream *consequence* of DEF-cluster haploinsufficiency — see Sections 8 and 11 — but infections do not cause the syndrome.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. A mother carries the **8p23.1 paracentric inversion polymorphism** in the heterozygous state (present in ~26% of Europeans) → predisposes to meiotic misalignment.
2. During **maternal meiosis**, the OR gene clusters (REPD/REPP) flanking 8p23.1 misalign → **NAHR** between the repeats.
3. NAHR **results in** a **dicentric chromosome intermediate** (and a reciprocal acentric fragment).
4. The dicentric **breaks** asymmetrically → **leads to** a chromosome bearing a terminal 8p deletion + an inverted interstitial 8p duplication.
5. The broken terminal end is **stabilized by telomere healing** (direct telomeric-repeat addition) or, rarely, telomere capture from 8q → yields a stable derivative chromosome 8.
6. Fertilization transmits the derivative chromosome → the zygote has **terminal 8p23 haploinsufficiency** + **interstitial 8p triplosensitivity**.
7. **Haploinsufficiency of GATA4/SOX7/NEIL2** (deletion) → impaired cardiac transcription-factor dosage → **results in** congenital heart defects and (with diaphragm-expressed genes) diaphragmatic hernia. *(Demonstrated in model organisms.)*
8. **Haploinsufficiency of the DEF cluster** (deletion) → reduced NK-cell activity and α-defensin → **leads to** immune vulnerability and severe respiratory infections. *(Demonstrated in one patient.)*
9. **Combined dosage imbalance of many 8p genes** (deletion and duplication) → disrupted neurodevelopment → **results in** corpus callosum anomalies, intellectual disability, hypotonia, and seizures. *(Gene-specific causality largely inferred; corpus-callosum-anomaly risk region mapped.)*
10. Structural malformations + neurodevelopmental impairment → **lead to** the lifelong multisystem clinical phenotype.

Detail by category

- **Molecular pathways / cellular processes.** The cardiac branch converges on **cardiac transcription-factor networks** (GATA4–TBX5–NKX2-5–HAND2; GATA4–GATA5–GATA6 combinatorial regulation) governing cardiomyocyte proliferation and septation. In mouse models, reduced Gata4 dosage lowers cardiomyocyte proliferation (Cdk4/Cdk2 downregulation; p21/Cdkn1a de-repression via FOG-2/NuRD). The vascular branch involves **SoxF–Notch** signaling (Sox7 upstream of hey2, efnb2, Dll4/Notch1) in arterial specification.

- GO suggestions: heart development (GO:0007507), cardiac septum morphogenesis (GO:0003279), regulation of cardiomyocyte proliferation, arterial endothelial cell differentiation (GO:0060842), corpus callosum morphogenesis / axon guidance, forebrain development.
- **Protein dysfunction.** Loss-of-dosage (haploinsufficiency) of GATA4/SOX7 and gain-of-dosage of interstitial genes; the mechanism is quantitative (gene dosage), not a mutant-protein misfolding mechanism.
- **Immune involvement.** DEF-cluster haploinsufficiency → reduced NK activity/α-defensin → impaired innate antiviral defense.
- **Cell types (CL suggestions):** cardiomyocyte (CL:0000746), endocardial/endothelial cell (CL:0002350/CL:0000115), arterial endothelial cell (CL:1000413), neuron (CL:0000540), natural killer cell (CL:0000623), skeletal muscle myocyte (hypotonia; CL:0000188).

Upstream mechanisms are the meiotic NAHR event and the resulting dosage imbalance; downstream are the organ-specific developmental failures (heart, brain commissures, diaphragm, immune cells).

Section 7 — Anatomical Structures Affected

Organ / body-system level. - **Central nervous system** (primary): corpus callosum (agenesis/hypoplasia), cerebral ventricles (ventriculomegaly), brain generally — UBERON:0002336 (corpus callosum), UBERON:0000955 (brain), UBERON:0002285 (lateral ventricle). - **Cardiovascular system** (primary): heart septa and valves, outflow tract, great vessels — UBERON:0000948 (heart), UBERON:0002099 (cardiac septum). - **Musculoskeletal system:** skeletal muscle (hypotonia), spine (scoliosis), limbs/joints — UBERON:0001134 (skeletal muscle), UBERON:0001130 (vertebral column). - **Craniofacial:** dysmorphic facial structures — UBERON:0001456 (face). - **Diaphragm** (secondary, subset): congenital diaphragmatic hernia — UBERON:0001103 (diaphragm). - **Renal** (secondary, prenatal): kidney anomalies — UBERON:0002113 (kidney). - **Immune system** (secondary): NK-cell function — UBERON:0002405.

Tissue / cell level. Nervous tissue (callosal projection neurons, glia), cardiac muscle (cardiomyocytes) and endocardium/endothelium, skeletal muscle, NK cells.

Subcellular level (GO CC). Nucleus (transcription-factor localization; GO:0005634) is central given the dosage effect on nuclear transcription factors (GATA4, SOX7). No specific mitochondrial/ER/lysosomal defect is established.

Localization / lateralization. CNS and cardiac malformations are typically **midline/bilateral** (the corpus callosum is a midline commissure; septal defects are central). Rare **laterality defects** (dextrocardia with corpus callosum agenesis) are reported (PMID: 20880309).

Section 8 — Temporal Development

- **Onset:** Congenital — the chromosomal imbalance is present from conception; malformations are prenatal/neonatal. Prenatal detection is possible via ultrasound anomalies.
- **Onset pattern:** Chronic/congenital (structural anomalies fixed at birth); seizures have a later, childhood onset (mean 3.9 y, range 2 months–9 years).
- **Progression:** Structural anomalies are **stable/non-progressive**; the disorder is a **chronic, lifelong** condition. Developmental delay is persistent; seizures may be episodic and require ongoing management.
- **Disease course:** Non-remitting congenital disorder; no spontaneous remission of core features. Severity is set largely at conception by rearrangement extent (and any mosaicism).
- **Critical periods:** Embryonic organogenesis (cardiac septation, diaphragm formation ~E11.5–12.5 equivalent, commissural/corpus-callosum development) is the window in which the dosage imbalance produces malformations; the principal postnatal intervention window is **early developmental intervention** and surgical correction of malformations.

Section 9 — Inheritance and Population

Epidemiology. inv dup del(8p) is **rare with no established population prevalence** (Orphanet lists it among rare chromosomal anomalies). For scale, the reciprocal isolated 8p23.1 duplication syndrome has an estimated prevalence of ~1/58,000. In an unselected pediatric developmental-disorder cohort, pathogenic 8p CNVs occurred in ~1% (10/966), of which inv dup del(8p) is a subset.

> "found 10 individuals with pathogenic copy number variants (CNVs) on the short arm of chromosome 8 (8p), representing approximately 1% of the patients analyzed" — PMID: 20461109

Inheritance. De novo, sporadic, arising in **maternal meiosis**. Parental karyotypes are typically normal. The predisposing factor is maternal heterozygosity for the benign 8p23.1 inversion (~26% of Europeans). Recurrence risk is low. Rare somatic-mosaic forms occur postzygotically.

> "Since inv dup(8p)s originate consistently in maternal meiosis, we investigated the maternal chromosomes 8 in eight mothers of subjects with inv dup(8p) ... All the mothers were heterozygous for an 8p submicroscopic inversion that was delimited by the 8p-OR gene clusters and was present, in heterozygous state, in 26% of a population of European descent." — PMID: 11231899

Penetrance / expressivity. Penetrance is high for the classic rearrangement; **expressivity is highly variable**, scaling with imbalance size/breakpoints and attenuated by mosaicism (below). No genetic anticipation (not a repeat-expansion disorder). Founder effects and consanguinity are not relevant. "Carrier frequency" in the classic sense does not apply; the relevant population parameter is the ~26% maternal inversion-polymorphism frequency (a susceptibility, not a disease-carrier state).

Demographics. Both sexes affected (autosomal; no strong sex bias reported). The inversion polymorphism is documented at ~26% in European-descent populations; the operative maternal factor is inversion-carrier status, not maternal age. The 8p23.1 segmental-duplication architecture is also a general genomic-instability hotspot (e.g., somatic 8p loss in ~31% of multiple myeloma).

Section 10 — Diagnostics

Definitive diagnosis is molecular-cytogenetic. - **Chromosomal microarray (CMA; aCGH or SNP-array):** first-line; defines precise breakpoints and the characteristic pattern — terminal 8p deletion + interstitial inverted duplication separated by a single-copy disomic region. - **G-banded karyotype:** typically shows add(8)(p23) or der(8). - **Metaphase FISH:** confirms the inverted duplication and 8p subtelomere deletion; subtelomeric/centromeric/whole-chromosome painting probes. - **Parental karyotyping:** documents de novo origin and can reveal the maternal 8p23.1 inversion.

> "aCGH detected an 11.35 Mb deletion in 8p23.3-p23.1 encompassing SOX7 and GATA4, and a 31.99 Mb duplication in 8p23.1-p11.1 in the fetus. Metaphase FISH confirmed inv dup del(8p)." — PMID: 27343326

Prenatal diagnosis. Invasive testing (CVS/amniocentesis) with CMA is prompted by ultrasound findings — increased NT, ventriculomegaly, cardiac defects, renal/craniofacial anomalies — that can mimic trisomy 18. In prenatal SNP-array cohorts, multisystem ultrasound anomalies confer the highest CMA yield (~27%) and increased NT is the strongest soft-marker predictor of chromosomal pathology.

> "multisystem anomalies conferred the highest risk (27.3%), driven predominantly by aneuploidies; among soft markers, increased nuchal translucency (NT) emerged as the strongest predictor of chromosomal pathology" — PMID: 42067806

Supporting clinical work-up. Brain **MRI** (corpus callosum anomalies, ventriculomegaly); **echocardiography** (septal/valve defects); orthopedic/skeletal evaluation; immune work-up (NK-cell activity, α-defensin) where infections recur. **Prenatal WGS** and low-coverage WGS can also detect the large CNVs with performance comparable to CMA.

Differential diagnosis. Trisomy 18 (prenatal overlap), isolated 8p23.1 deletion or duplication syndromes, other contiguous-gene syndromes with corpus callosum agenesis and CHD, Kabuki-like phenotypes; distinguished by the characteristic dual deletion+duplication CMA signature.

Screening. No population/newborn screening exists for inv dup del(8p); detection is via diagnostic (not screening) CMA prompted by phenotype, or prenatally by ultrasound-triggered testing.

Section 11 — Outcome / Prognosis

Nature. Chronic, lifelong congenital disorder; **no cure**; management supportive.

Mortality. Prognosis is dominated by malformation severity. **Congenital heart defects** (including hypoplastic left heart) and **congenital diaphragmatic hernia** are the principal life-threatening complications; neonatal death occurs with severe cardiac malformation (e.g., severe polyvalvular dysplasia, death at day 12).

> "the present case had severe polyvalvular dysplasia and the infant deceased at day 12 of life" — PMID: 28211984
>
> "Recurrent interstitial deletion of a region of 8p23.1 flanked by the low copy repeats 8p-OR-REPD and 8p-OR-REPP is associated with a spectrum of anomalies that can include congenital heart malformations and congenital diaphragmatic hernia (CDH). Haploinsufficiency of GATA4 is thought to play a critical role in the development of these birth defects." — PMID: 19606479

Morbidity. Survivors have persistent intellectual disability, hypotonia, orthopedic problems (scoliosis, joint/limb anomalies), and require lifelong developmental support. **Immune vulnerability** (DEF haploinsufficiency) predisposes to severe respiratory infections (severe RSV bronchiolitis; a severe COVID-19 case requiring 26-day hospitalization with 9 days in PICU and mechanical ventilation).

> "This patient showed lower NK cell activity and α-defensin level compared with healthy controls." — PMID: 35768224

Prognostic factors. Rearrangement size/breakpoints, presence and severity of CHD/CDH, and mosaicism. **Milder outcomes** occur with smaller duplications, isolated terminal deletions distal to GATA4, or somatic mosaicism.

> "This female has developmental delay, but lacks congenital anomalies that are associated with either 8p abnormality in non-mosaic form. The attenuated phenotype in this individual may be due to compensation of one cell line for imbalances in the other cell line." — PMID: 20830805
>
> "unlike the inv dup del(8p), the phenotype in our case is milder with no central nervous system malformations or cardiac defects" — PMID: 18302246

Section 12 — Treatment

No disease-specific or curative therapy exists. There are no pharmacological, gene, or cell therapies targeting the rearrangement, and no registered disease-specific clinical trials. Management is **supportive and multidisciplinary**:

Domain	Intervention	NCIT suggestion
Development	Early developmental intervention; physical, occupational, speech therapy	Rehabilitation Therapy (NCIT:C15917)
Cardiac	Surgical repair of congenital heart defects	Cardiac Surgery (NCIT:C157664)
Diaphragm	Surgical repair of diaphragmatic hernia	Surgical Procedure (NCIT:C15329)
Epilepsy (~34%)	Anti-seizure medication	Anticonvulsant Agent (NCIT:C264)
Orthopedic	Management of scoliosis/limb/joint anomalies	Orthopedic Procedure
Nutrition/ feeding	Feeding support	Nutritional Support (NCIT:C15311)
Infections	Intensive/critical care (mechanical ventilation, dexamethasone, remdesivir in severe COVID-19)	Supportive Care (NCIT:C15277)

> "There, she was mechanically ventilated, received dexamethasone and remdesivir, and was hospitalized for 26 days, nine of which were in the pediatric intensive care unit." — PMID: 37829974

Pharmacogenomics, advanced therapeutics, targeted/immunotherapy, personalized medicine: none applicable/available for this disorder.

Section 13 — Prevention

- **Primary prevention:** Not possible (de novo meiotic event). No vaccination or risk-factor modification prevents the rearrangement.
- **Secondary prevention: Prenatal diagnosis** by CMA (prompted by ultrasound anomalies/ increased NT) enables informed reproductive decisions and delivery planning. Preimplantation genetic testing is an option for known inversion carriers.
- **Tertiary prevention:** Prevent complications — surgical correction of malformations, seizure control, and **infection prophylaxis** (COVID-19 vaccination and RSV prophylaxis are advised given the DEF-related immune vulnerability).
- **Genetic counseling:** Central preventive measure. Because the event arises de novo from a common benign maternal inversion polymorphism, recurrence risk is low, but the same NAHR mechanism can generate reciprocal deletion/duplication products, so prenatal monitoring is recommended.

> "Prenatal diagnosis should be performed to monitor the recurrent risk of inv dup del(8p), as well as the other three harmful consequences resulted from the same NAHR mechanism." — PMID: 20677137

Section 14 — Other Species / Natural Disease

There is **no naturally occurring animal disease** equivalent to inv dup del(8p) — the specific human 8p23.1 OR-cluster/inversion architecture and NAHR-driven rearrangement are human-specific. However, the **individual dosage-sensitive genes are evolutionarily conserved** and modeled experimentally (Section 15): - **Mouse *Gata4*** (NCBI Gene 14463) — ortholog of human GATA4. - **Zebrafish *sox7, sox18*** — orthologs of the human SOXF family. - Taxonomy: *Mus musculus* (NCBI:txid10090), *Danio rerio* (NCBI:txid7955).

No zoonotic potential or cross-species transmission (non-infectious genetic disorder). No breed-specific (VBO) associations.

Section 15 — Model Organisms

No single model reproduces the entire inv dup del(8p) contiguous imbalance, but **gene-specific models recapitulate the cardiac and vascular components**, validating the dosage mechanism.

Mouse (mammalian): - *Gata4* is a core cardiac transcription factor; disruption causes congenital heart defects. **Gata4(+/-);Tbx5(+/-)** embryos show decreased atrial/ventricular myocardial thickness and atrioventricular septation defects with reduced cardiomyocyte proliferation (Cdk4/Cdk2 downregulation). - **Gata4/Gata5** compound heterozygotes develop double-outlet right ventricle, VSDs, and valve defects.

> "Gata4(+/-);Tbx5(+/-) mouse embryos display decreased atrial and ventricular myocardial thickness at E11.5, prior to cardiac septation." — PMID: 24858909

Zebrafish: - *sox7* mutants show a "short circulatory loop" from aberrant artery–vein connections; Sox7 acts upstream of Notch (hey2, efnb2) in arterial specification. Combined Sox7/Sox18 loss ablates the dorsal aorta.

> "sox7 mutants display a short circulatory loop around the heart as a result of aberrant connections between the lateral dorsal aorta (LDA) and either the venous primary head sinus (PHS) or the common cardinal vein (CCV)" — PMID: 25834021

Model characteristics. These models faithfully reproduce the **cardiac septal/outflow and vascular defects** of the 8p23.1-deletion component. **Limitations:** they capture only single-gene dosage effects, not the combined deletion+duplication imbalance, the neurodevelopmental/corpus-callosum phenotype, or the immune (defensin) component. **Resources:** MGI (mouse *Gata4*), ZFIN (zebrafish *sox7*/*sox18*).

Mechanistic Model (Synthesis)

```
Maternal 8p23.1 inversion polymorphism (heterozygous, ~26% Europeans)
                    │  predisposes to
                    ▼
   Meiotic misalignment of OR repeats (REPD/REPP) at 8p23.1
                    │  NAHR
                    ▼
        Dicentric chromosome 8 intermediate
                    │  asymmetric breakage
                    ▼
  der(8): TERMINAL 8p DELETION  +  INVERTED INTERSTITIAL 8p DUPLICATION
                    │  stabilized by telomere healing / (rare) 8q capture
                    ▼
       ┌──────────────┴───────────────┬───────────────────────┐
       ▼                              ▼                       ▼
  GATA4/SOX7/NEIL2 loss        DEF cluster loss      Global 8p dosage imbalance
  (deletion, dosage↓)          (deletion)            (deletion + duplication)
       │                              │                       │
       ▼                              ▼                       ▼
  Congenital heart defects     ↓NK activity,         Corpus callosum anomalies,
  ± diaphragmatic hernia       ↓α-defensin           ID, hypotonia, seizures,
  [model-validated]            → severe infections    dysmorphism [largely inferred]
       └──────────────┬───────────────┴───────────────────────┘
                    ▼
        Lifelong multisystem clinical phenotype
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Upstream = the maternal inversion + NAHR + dosage imbalance. **Downstream** = organ-specific developmental failures. The cardiac and vascular branches are experimentally demonstrated (mouse/zebrafish); the neurodevelopmental branch is well correlated but gene-level causality remains largely inferred.

Evidence Base — Key Literature

PMID	Contribution	Evidence type
24502041	NAHR/maternal-meiosis/OR-cluster mechanism; 26% inversion frequency	Human, cytogenetic
11231899	Landmark: maternal meiotic origin; all mothers inversion carriers	Human, cytogenetic
19041960	Telomere healing stabilizes terminal deletion; 8q telomere capture	Human, case
34866188	Largest cohort (36 pts): ID 97%, seizures 34%, AnCC 63%	Human, cohort
35327368	8-case series: psychomotor/language delay, hypotonia, dysmorphism, CC agenesis	Human, series
23165946	GATA4/NEIL2/SOX7 → diaphragmatic (and cardiac) defects	Human + network
35768224	DEF haploinsufficiency → low NK activity/α-defensin → severe RSV	Human, case
26097203	Reciprocal 8p23.1 duplication prevalence ~1/58,000	Human, cohort
28533195	8p23.1 deletion syndrome spectrum (CHD, ID, behavior, microcephaly, epilepsy)	Human, review/ case
18393291	GATA4 as causal gene for cardiac phenotype	Human, mapping
24858909	Mouse Gata4/Tbx5: myocardial/septation defects	Mouse model
25834021	Zebrafish sox7: cardiovascular defects, Sox7 upstream of Notch	Zebrafish model
27343326	aCGH + FISH diagnostic workflow (SOX7/GATA4 deletion)	Human, prenatal
42067806	Prenatal CMA yield; NT strongest predictor	Human, cohort
28211984	Neonatal mortality from severe polyvalvular dysplasia	Human, case
19606479	8p23.1 deletion → CHD + CDH; GATA4 haploinsufficiency critical	Human, series
37829974	Severe COVID-19 in affected infant; intensive care	Human, case
20677137	Prenatal diagnosis/counseling as prevention	Human, case
20830805	Mosaicism attenuates phenotype	Human, case

PMID	Contribution	Evidence type
18302246	Tandem vs inverted duplication modulates severity	Human, case
20461109	Pathogenic 8p CNVs ~1% of developmental-disorder cohort	Human, cohort
20880309	Dextrocardia + corpus callosum agenesis (laterality)	Human, case

Limitations and Knowledge Gaps

1. **No prevalence estimate.** The disorder's true incidence/prevalence is unknown; evidence is from case reports and small series, biasing toward severe/recognizable cases.
2. **Neurodevelopmental gene-level causality is inferred, not demonstrated.** While the cardiac (GATA4/SOX7) and immune (DEF) branches are mechanistically supported by models/patient data, the specific genes responsible for corpus callosum anomalies, intellectual disability, and seizures within the large duplication/deletion intervals are not individually proven.
3. **No whole-syndrome animal model.** Existing models capture single-gene dosage effects only; the combined deletion+duplication imbalance and the neurodevelopmental phenotype are not modeled.
4. **Genotype–phenotype correlation is incomplete.** One 8-case series found no clear correlation between molecular-cytogenetic variants and clinical severity, though a corpus-callosum-anomaly risk region has been mapped.
5. **No natural-history/QoL data.** Longitudinal outcome, life-expectancy, and standardized quality-of-life data are lacking.
6. **Epigenetic and multi-omic characterization absent.** No transcriptomic, proteomic, metabolomic, or methylation profiling specific to inv dup del(8p) is available.

Proposed Follow-up Experiments / Actions

1. **Establish an international registry** to define prevalence, natural history, mortality, and QoL with standardized instruments.

2. **Breakpoint–phenotype mapping** across large CMA cohorts to refine the corpus-callosum, seizure, and ID critical regions and to test dosage of specific candidate genes (XKR6, MIR597, and duplication-interval genes).
3. **Neurodevelopmental modeling:** generate patient-derived iPSCs / cerebral organoids carrying the rearrangement (or engineered 8p dosage changes) to test corpus-callosum/ neuronal phenotypes and identify driver genes.
4. **Systematic immune phenotyping** (NK activity, α-defensin, infection history) across a patient cohort to determine how consistently DEF haploinsufficiency causes clinically relevant immunodeficiency and to guide prophylaxis (RSV, COVID-19 vaccination).
5. **Prospective cardiac outcome study** correlating GATA4/SOX7 deletion status with CHD type/severity and surgical outcomes.
6. **Transcriptomic/methylation profiling** of patient tissues (or organoids) to search for a diagnostic episignature and downstream dysregulated pathways.

Report compiled from 5 investigation iterations, 11 confirmed findings, and 53 primary papers. Evidence types are distinguished throughout as human clinical, model organism, in vitro, or computational.
